

Airbnb Dynamic Pricing Recommendation Engine

Introduction

Airbnb hosts need competitive pricing to maximize occupancy and revenue. This project analyzes the AB_NYC_2019 dataset to identify the factors affecting Airbnb listing prices and to develop a basic price prediction model.

Abstract

The project uses Python for data cleaning, exploratory data analysis, visualization, and regression modeling. Tableau Public is used to build an interactive dashboard for pricing insights.

Tools Used

Google Colab, Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Tableau Public, Microsoft Excel (CSV dataset).

Steps Involved

1. Imported the dataset.
2. Cleaned missing values and duplicates.
3. Performed EDA and visualizations.
4. Built Linear Regression and Random Forest models.
5. Evaluated using MAE, RMSE and R^2 .
6. Created a Tableau dashboard with interactive filters.

Conclusion

The analysis identified neighbourhood group, room type, reviews, and availability as important pricing factors. The prediction model and dashboard can assist hosts in making better pricing decisions.